

FOREMOST LOAD CELL CALIBRATION

A. LOAD CELL

The weights cannot be calibrated when the draft weigher is running.

If you access the calibration screen when the draft weigher is active, the button will not be displayed

NOTE: power should be on for at least (15) minutes before starting the calibration Procedure

To calibrate the draft weigher:

- 1) Empty the chamber that you wish to calibrate.
- 2) Enter the calibration weight that will be used.
- 3) Press the zero button.

The current weight displayed should now read 0.0

- 4) Put the calibration weight on the chamber.
- 5) After waiting a few seconds to let the chamber settle, press the calibration button.

The current weight displayed should now agree with the calibration weight.

- 6) Remove the weights from the chamber. The current weight displayed should return to 0.0. If it does not, then repeat steps 3-6.

The chamber is now calibrated

The minimum and maximum weights are calculated when the chamber is calibrated. If after calibrating a chamber the minimum weight is greater than -1.00 or the maximum weight is less than the preset limit a message will be displayed. This indicates that either the calibration was done incorrectly or the pots on the analog signal conditioner need to be adjusted. If after adjusting the pots, the minimum or maximum is still displayed, then the chamber must be checked to see if the load cell has been damaged or stressed beyond the limits of the signal conditioner.

B. Adjusting the analog signal conditioner potentiometers

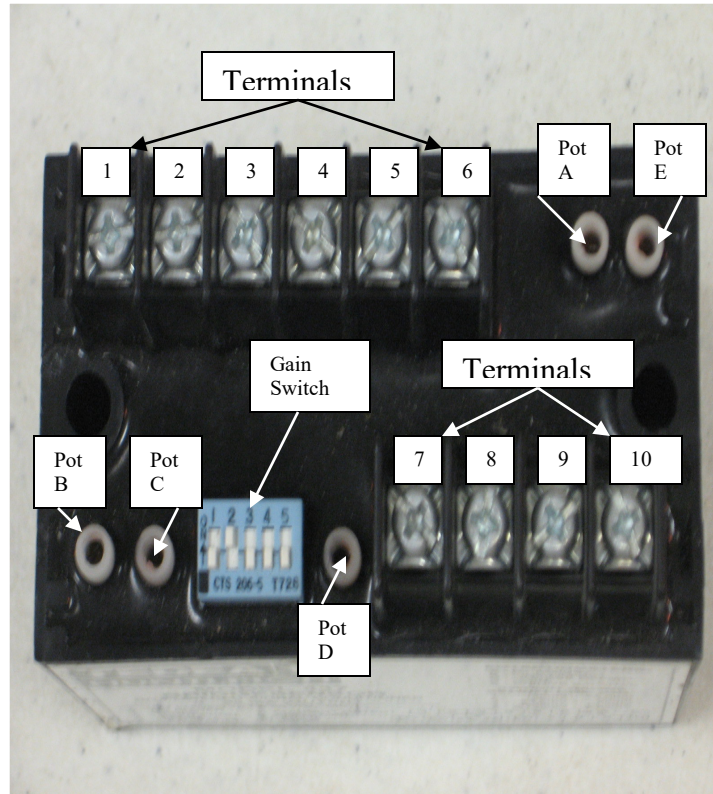
- 1) POT (B) FINE OFFSET AND POT (C) COARSE OFFSET.

Adjusting these pots clockwise, (and pressing the zero button on the calibration screen), will Cause the minimum reading displayed to increase. That is, it will become more negative. The Maximum reading will decrease.

	Current weight	Minimum reading
Before turning pot -	0.00	-1.00
After turning pot -	1.00	-1.00
After setting zero -	0.00	-2.00

After adjusting these pots counter-clockwise, (and pressing on the zero button on the Calibration screen), will cause the minimum reading displayed to decrease. That is, it Will become less negative. The maximum reading will increase.

	Current weight	Minimum reading
Before turning pot -	0.00	-5.00
After turning pot -	-3.00	-5.00
After setting zero -	0.00	-2.00



2) POT (D) GAIN

Adjusting these pots clockwise, (and then pressing the calibration button on the calibration Screen), will cause the maximum reading displayed to decrease. That is, it will become smaller. The minimum reading displayed will also decrease.

(I.E. Before – 35.54 After -30.54)

Adjusting these pots counter-clockwise, (and then pressing on the calibration button on the Calibration screen), will cause the maximum reading display to increase, That is, it will become larger. The minimum reading displayed will also increase.

(I.E. Before – 25.54 After – 30.54)

C. Signal switch selection

Note: This signal conditioner does not normally need to be recalibrated unless the load cell or the Unit is changed

- 1) Connect a D.C. Meter (millivolt scale) to pins 4&5. Adjust pot (A) until you read 10 VDC.
- 2) With weight chamber empty, connect a D.C. Meter (millivolt scale) to pins 9&10. Adjust pot (E) until you read 0.00 mv
- 3) Apply a known weight to weight chamber and read the millivolt signal.
- 4) Take the difference between each reading, (and using the formula below), set the gain switch to

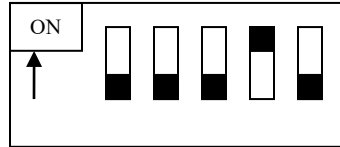
the expected load cell output. (see table below and example).

FORMULA: $\frac{(\text{DIFF.}) (\text{MAX. READING DESIRED})}{\text{KNOWN WEIGHT}} = \text{EXPECTED OUTPUT}$

Table 1- Model 437 gain switch

Only one switch on for expected transducer full scale output

Bridge output signal for: 0 to 20ma ma



Gain switch position "ON":	1	50.0mv	to	62.5mv
	2	25.0mv	to	50.0mv
	3	12.5mv	to	25.5mv
	4	6.25mv	to	12.5mv
	5	3.125mv	to	6.25mv

Example: We know the dead load reading (empty) is 10.00mv and the known weight reading (full) is 9.3mv. select proper gain switch. Then using a maximum reading desired of 100 lbs. and a known weight of 50 lbs., then;

$$\text{Expected output} = \frac{(9.3) (100)}{50} = 18.6\text{mv}$$

The answer is switch #3, because switch #3 range is from 12.5mv to 25.0mv